

Project Design Phase
Solution Architecture

Date	8 July 2026
Team ID	SWTID-2026-5356
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is the process of designing an effective system that transforms agricultural data into meaningful insights for decision-making. It connects the agricultural crop production dataset with data processing, visualization, and web technologies to provide an easy-to-use analytics platform. Its goals are to:

- Analyse agricultural crop production data from different states, districts, seasons, and years to identify production patterns and trends.
- Design a structured analytics system that includes data cleaning, preparation, interactive visualization, and web application integration for stakeholders.
- Provide interactive dashboards and reports that help farmers, researchers, and government agencies make informed decisions based on agricultural data.

Solution Architecture Diagram:

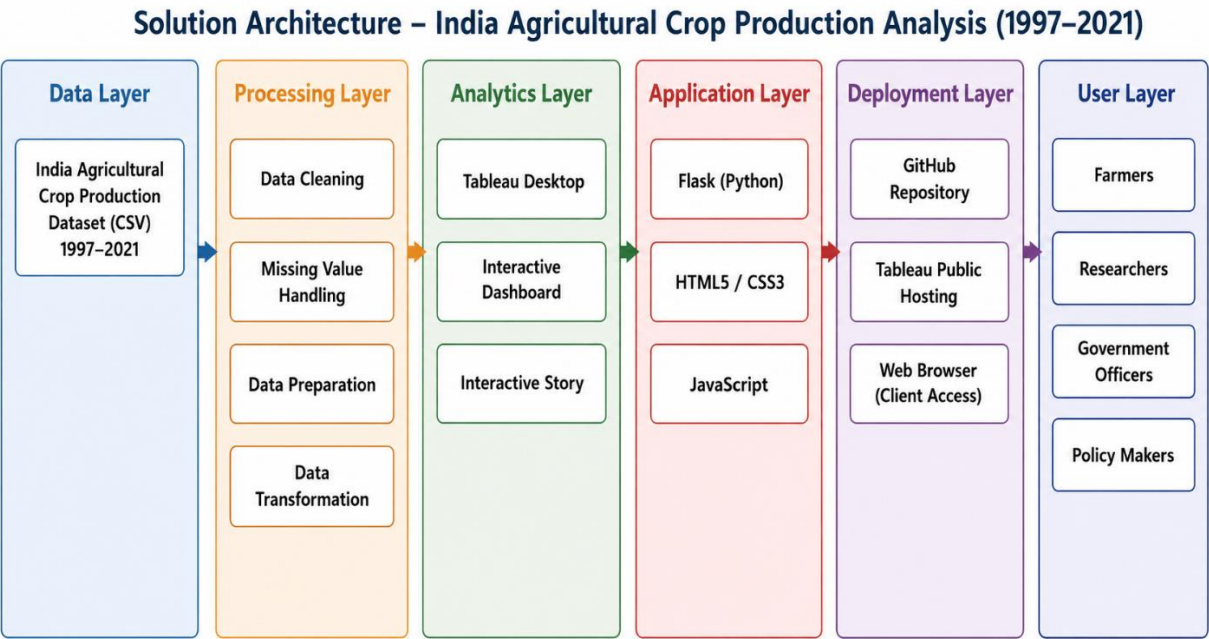


Figure 1: Architecture and data flow of the India Agricultural Crop Production Analysis (1997–2021) application